CS201 Composite

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***Abstract*— Write only ONE function per question in C++. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can use any function given in the data structure.**

I. Question (Algorithm Analysis), 20 points

```
int algorithm(int N){
  int i, j, k, sum = 0;
  for (i = 1; i <= N; i++)
      for (j = i; j <= N - i; j++)
          for (k = 1; k <= j - i; k++)
             if ( (j+k) %2 == 0)
                  sum++;
  return sum;
}
```

What does the function algorithm return in terms of N? Assume that N is even. Show all your work.

II. Question (Linked List), 20 points

Write method

`bool containsOnlyTriplicates()`

which returns true if the singly linked list only contains triplicates, that is, every datum (number) occurs only three times. Important warning, the triplicate elements may not be adjacent. You are not allowed to use any singly linked list methods, just attributes, constructors, getters and setters.

III. Question (Graph), 20 points

A graph is 2-colorable if each vertex can be given one of 2 colors, and no edge connects identically colored vertices. Write linear-time method

`bool isTwoColorable()`

to test a graph for 2-colorability of an undirected graph by using breadth first search where graph is stored in adjacency list format. You may assume graph is connected. You may only use array as an extra data structure.

IV. Question (Tree), 20 points

Write a nonrecursive method using Stack

`int leftistOrRightist()`

that finds the difference between the number of leftist nodes and rightist nodes in a binary search tree. A node is leftist (rightist) if it has only left (right) child.

V. Question (Sorting), 20 points

Suppose you are given an array of N integers. Write a single pass algorithm

`void quickTriple(int* A, int N)`

that puts the numbers fully divisible by 3 first, then the numbers which division remains 1, and lastly the numbers which division remains 2 similar to the partition algorithm in QuickSort.